

PROJECT REPORT

Streamlining IT Procurement: Automating Standard Laptop Orders with Flow

IT Procurement Workflow Automation Project

Project Information	Details
Project Title	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow
Domain	IT Procurement / Workflow Automation
Automation Platform	Flow-based workflow automation
Project Type	Business Process Automation
Team Members	[Enter team member names and roles]
Guide / Mentor	[Enter guide name]
Institution / Organization	[Enter institution / organization]
Academic Year	2026–2027

Project Report Format

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1. INTRODUCTION

1.1 Project Overview

The project “Streamlining IT Procurement: Automating Standard Laptop Orders with Flow” focuses on automating the repetitive activities involved in requesting and procuring standard laptops. The workflow captures a laptop request, validates required information, checks whether the request matches an approved standard configuration, routes the request through the required approval path, and creates or updates the procurement record automatically.

The objective is to replace manual email-based coordination and spreadsheet tracking with a consistent, traceable and rule-driven process. Flow acts as the orchestration layer connecting the requester, manager, IT/procurement teams and the procurement record.

1.2 Purpose

The purpose of the project is to reduce procurement turnaround time, minimize manual data entry, improve visibility of order status and enforce standard laptop purchasing rules.

- Standardize the laptop request process.
- Automatically validate request details and eligibility.
- Route requests to the correct approver.
- Create procurement tasks/orders without duplicate manual entry.
- Notify stakeholders at important workflow stages.

- Maintain an auditable history of decisions and status changes.

2. IDEATION PHASE

2.1 Problem Statement

Employees who need a standard laptop may currently depend on email, chat messages, spreadsheets or separate procurement forms. This can result in incomplete requests, repeated follow-ups, delayed approvals, inconsistent device selection and limited visibility into the order lifecycle.

The problem addressed by this project is: How can a standard laptop order be processed automatically from request submission to procurement completion while preserving approval controls and status visibility?

2.2 Empathy Map

Perspective	Key Observations
Requester – Says	“I need a laptop quickly and want to know when it will arrive.”
Requester – Thinks	The process should be simple and should not require repeated data entry.
Requester – Does	Submits a request, waits for approval, and follows up for status.
Requester – Feels	Frustrated when the request is delayed or status is unclear.
IT / Procurement	Needs complete data, approved configurations, clear ownership and an audit trail.
Manager / Approver	Needs concise information to make a quick approval decision.

2.3 Brainstorming

- Create a single standard laptop request form.
- Use rules to distinguish standard requests from exceptions.
- Automatically assign approvals based on request attributes.
- Generate procurement tasks or orders automatically after approval.
- Send confirmation and status notifications automatically.
- Escalate overdue approvals or procurement tasks.
- Record timestamps and decisions for reporting and audit.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	User Action	System / Flow Action	Expected Outcome
Request	Employee submits laptop request.	Validate mandatory fields.	Complete request is created.
Validation	Request waits for processing.	Check standard model, budget and eligibility rules.	Eligible request proceeds automatically.
Approval	Manager reviews request.	Send approval task and track response.	Approved or rejected with reason.
Procurement	Procurement team processes order.	Create/update procurement task or purchase record.	Order enters procurement queue.
Fulfilment	IT prepares device.	Update status and notify requester.	Requester receives fulfilment updates.
Closure	Device is delivered.	Close request and retain audit history.	Completed procurement record.

3.2 Solution Requirements

Functional Requirements

- The system shall provide a standard laptop request form.
- The system shall validate mandatory requester, department, delivery and laptop configuration fields.
- The Flow shall identify whether the requested configuration is part of the approved standard catalogue.
- The Flow shall route requests for approval according to defined business rules.
- Approved requests shall create or update the procurement record automatically.
- The Flow shall notify the requester and relevant teams when the status changes.
- Rejected requests shall record a rejection reason.
- The system shall maintain timestamps and an audit trail for workflow actions.

Non-Functional Requirements

- Reliability: workflow steps should execute consistently without duplicate order creation.
- Security: only authorized users should submit, approve or administer requests.
- Performance: standard requests should progress through automated steps with minimal processing delay.
- Maintainability: business rules and standard catalogue values should be easy to update.
- Usability: requesters should complete the form with minimal training.
- Auditability: all key approvals, status changes and exceptions should be traceable.

3.3 Data Flow Diagram

The following logical data flow can be used as the basis for the implemented Flow:

Step	Flow Component	Data / Action
1	Requester	Submit laptop request
2	Request Form	Capture employee, department, location, model and justification
3	Validation	Check mandatory fields and standard catalogue
4	Approval	Route to manager / designated approver
5	Procurement	Create procurement task or purchase record
6	Notification	Send status updates to requester and stakeholders
7	Fulfilment	IT prepares and assigns the laptop
8	Closure	Mark request complete and store audit history

3.4 Technology / Automation Stack

Layer	Technology / Component	Purpose
User Interface	Request form / portal	Collect standard laptop requirements
Workflow	Flow / workflow engine	Automate rules, approvals, tasks and notifications
Data	Request, employee, catalogue and procurement records	Store workflow information
Business Rules	Conditional logic / decision rules	Validate standard configuration and routing
Notifications	Email / in-platform notifications	Communicate workflow progress
Reporting	Workflow reports / dashboards	Monitor volume, status and turnaround time

4. PROJECT DESIGN

4.1 Problem–Solution Fit

Current Challenge	Proposed Flow Capability
Requests arrive through multiple channels.	Use a single standardized request entry point.

Incomplete information causes rework.	Mandatory fields and validation rules.
Approvals are manually chased.	Automatic approval routing and reminders.
Procurement staff re-enter request details.	Create procurement records from approved request data.
Requester cannot easily see status.	Automatic status notifications and tracking.
Exceptions are difficult to identify.	Conditional branching and exception queues.

4.2 Proposed Solution

The proposed solution is a rule-driven Flow that starts when a standard laptop request is submitted. The Flow validates the request, checks the standard catalogue and approval conditions, routes the request for approval, and creates the procurement work item when approval is granted. Notifications are triggered at submission, approval, procurement and completion stages.

If the request does not meet standard rules—for example, a non-standard model or a value above the defined threshold—the Flow routes it to an exception path instead of creating a normal standard order.

4.3 Solution Architecture

Logical architecture:

Component	Responsibility	Interaction
Requester	Initiates standard laptop request	Submits form to workflow
Request Record	Stores request details and status	Read/write by Flow
Flow Engine	Controls validation, decisions and routing	Reads request and executes actions
Approval Layer	Manager or authorized approver	Approve / reject
Procurement Record	Tracks purchase processing	Created/updated after approval
IT Fulfilment	Prepares and assigns device	Updates fulfilment status
Notification Service	Communicates status	Sends automated messages

High-level flow: Requester → Request Form → Validation → Standard Catalogue Check → Approval → Procurement Record → IT Fulfilment → Closure → Notification.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Phase	Activities	Deliverable	Suggested Duration
1. Discovery	Understand current procurement process and stakeholders.	Problem definition	1 week

2. Requirements	Define fields, rules, approvals and exceptions.	Requirement specification	1 week
3. Design	Design Flow logic, data mapping and notifications.	Workflow design	1 week
4. Development	Build request form, Flow and procurement actions.	Working automation	2 weeks
5. Testing	Execute functional, negative and performance tests.	Test report	1 week
6. Deployment	User validation, training and production release.	Production workflow	1 week
7. Review	Monitor results and tune rules.	Improvement backlog	Ongoing

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Test ID	Scenario	Expected Result	Status
TC-01	Submit complete standard laptop request	Request created and Flow starts.	Pass / Update
TC-02	Submit request with missing mandatory field	Submission blocked or validation error shown.	Pass / Update
TC-03	Standard configuration selected	Request follows standard procurement path.	Pass / Update
TC-04	Non-standard configuration selected	Request follows exception path.	Pass / Update
TC-05	Approver approves request	Procurement record is created/updated.	Pass / Update
TC-06	Approver rejects request	Request closes/reverts with rejection reason.	Pass / Update
TC-07	Duplicate trigger condition	No duplicate procurement order is created.	Pass / Update
TC-08	Notification event occurs	Correct stakeholder receives notification.	Pass / Update
TC-09	Procurement status changes	Requester receives updated status.	Pass / Update

TC-10	Workflow completes	Request is closed and audit history retained.	Pass / Update
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Performance measurements should be captured in the actual environment. Recommended measures include workflow execution time, approval turnaround time, procurement processing time, notification delay, error rate and duplicate-order rate. Replace the placeholders with measured values after execution.

7. RESULTS

7.1 Output Screenshots

Insert screenshots from the implemented Flow in this section. Recommended screenshots:

- Standard laptop request form.
- Flow trigger and initial validation step.
- Standard catalogue decision / condition branch.
- Approval action and approval notification.
- Procurement record generated by the Flow.
- Requester status notification.
- Completed request and audit/history view.
- Dashboard or report showing procurement status and turnaround time.

[Insert Screenshot Here]
Figure 1 – Laptop Request Form

[Insert Screenshot Here]
Figure 2 – Flow Logic / Decision Branch

[Insert Screenshot Here]
Figure 3 – Procurement Record

[Insert Screenshot Here]
Figure 4 – Completion / Status View

8. ADVANTAGES & DISADVANTAGES

Advantages	Disadvantages / Considerations
Reduces repetitive manual work.	Initial workflow design requires analysis and configuration.

Improves request consistency.	Poorly defined rules can route requests incorrectly.
Speeds up standard approvals and procurement.	Workflow failures require monitoring and support.
Provides better status visibility.	Changes to procurement policy may require rule updates.
Creates an audit trail.	Integration or notification dependencies can affect automation.
Reduces duplicate data entry.	Exception cases still need controlled manual handling.

9. CONCLUSION

The project demonstrates how a Flow-based automation can streamline standard laptop procurement by connecting request intake, validation, approvals, procurement processing, notifications and closure into one controlled workflow. The approach improves process consistency and visibility while reducing repetitive administrative effort.

The solution is designed so that standard requests can move through a predictable path, while non-standard requests are redirected to an exception process. This separation supports automation without removing necessary procurement controls.

10. FUTURE SCOPE

- Integrate directly with vendor catalogues and purchase-order systems.
- Add automatic stock and inventory availability checks before ordering.
- Introduce SLA timers, reminders and escalation for delayed approvals.
- Use analytics to measure procurement cycle time and identify bottlenecks.
- Add automated asset registration after laptop delivery.
- Support employee onboarding workflows that trigger laptop provisioning.
- Introduce policy-based approval thresholds that can be maintained without redesigning the entire Flow.
- Add exception analytics to identify frequently requested non-standard configurations.

11. APPENDIX

A. Standard Laptop Request Fields

Field	Example / Description	Required
Requester	Employee submitting the request	Yes
Employee ID	Unique employee identifier	Yes
Department	Requester's department	Yes
Location	Delivery / office location	Yes

Laptop Model	Approved standard model	Yes
Configuration	RAM, storage, processor, etc.	Yes
Business Justification	Reason for laptop request	Yes
Manager	Approving manager	Yes
Required By Date	Expected delivery date	Yes
Exception Reason	Required only for non-standard request	Conditional

B. Sample API / Integration Documentation

If the selected Flow platform exposes APIs or integrates with external procurement systems, document the actual endpoints used by the implementation. The following is a documentation template rather than a claim about a specific platform.

Operation	Purpose	Method / Interface	Authentication
Create Request	Create laptop procurement request	[Enter actual method/interface]	[Enter method]
Get Catalogue	Retrieve approved standard laptop models	[Enter actual method/interface]	[Enter method]
Create Procurement Record	Create purchase/procurement task	[Enter actual method/interface]	[Enter method]
Update Status	Update procurement / fulfilment status	[Enter actual method/interface]	[Enter method]
Send Notification	Notify requester or approver	[Enter actual method/interface]	[Enter method]

C. Authentication and Authorization

Access to the request, approval and procurement functions should be restricted according to organizational roles. Requesters should be able to submit and view their own requests; approvers should be able to act on assigned approvals; procurement and IT users should have access to the records required for fulfilment; administrators should manage workflow configuration. Document the actual authentication mechanism, role model, tokens or sessions used by the implemented platform.

D. Setup / Deployment Instructions

1. Create or configure the required request, catalogue, approval and procurement data structures.
2. Configure the standard laptop catalogue and business rules.
3. Build the request form and map each field to the request record.
4. Configure the Flow trigger and validation actions.

5. Configure standard versus exception decision branches.
6. Configure approval routing and notifications.
7. Configure procurement record creation or integration.
8. Run the test cases and record actual results.
9. Deploy the approved Flow to the production environment.

E. Source Code / Configuration (if any)

[Insert repository link, Flow export/package, configuration file reference, or state “Not applicable” if the solution is configured entirely through the Flow platform.]

F. Dataset / Reference Links

Dataset / catalogue reference: [Insert link or internal reference]

GitHub / Project Demo Link: [Insert link if applicable]

Streamlining IT Procurement: Automating Standard Laptop Orders with Flow

Salesforce Flow Automation • Functional, Performance & UAT Documentation

Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow
Platform	Salesforce Flow
Document Type	Project Development, Testing & UAT Documentation
Version	1.0
Date	01 October 2026

Prepared for academic / project submission

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Note: Page numbers can be refreshed after final editing in Word.

1. Project Overview

The project streamlines the procurement of standard laptops by replacing repetitive manual coordination with a Salesforce Flow. Employees submit a laptop request using a standardized request form. The Flow validates the request, determines whether the request meets the standard configuration, routes approval when required, creates the procurement record, and sends notifications to the relevant stakeholders.

The solution is designed for standard laptop orders where the organization has predefined device models, approved accessories, cost limits, and approval rules. Non-standard or incomplete requests are routed to an exception path instead of being automatically ordered.

Project Information

Parameter	Value
Automation Platform	Salesforce Flow
Primary Process	Standard Laptop Procurement
Primary Users	Employees, Managers, IT Procurement, IT Support
Trigger	New or submitted laptop request
Primary Output	Validated procurement request / order record
Notification	Email / Salesforce notification

2. Problem Statement

- Laptop procurement can involve repeated data entry, manual validation, approval follow-ups, and status communication.
- Manual processing increases the risk of incorrect device selection, missing information, duplicate requests, delayed approvals, and inconsistent records.
- The proposed Flow centralizes the business rules and automates the routine steps so that standard laptop requests can be processed consistently and traceably.

3. Objectives

Objective	Description
Standardize requests	Capture employee, department, location, device type, and justification in a consistent format.
Automate validation	Check mandatory fields, approved models, budget thresholds, and duplicate requests.
Automate approvals	Route requests to the appropriate manager or procurement approver based on defined rules.

Create records automatically	Create or update procurement/order records after successful validation and approval.
Improve visibility	Maintain status, timestamps, owner, and audit information in Salesforce.
Reduce manual effort	Use Flow to handle routine routing, notifications, and record updates.

4. Scope

In Scope

- Standard laptop request intake and validation.
- Approved laptop model and configuration validation.
- Budget / cost threshold checks.
- Duplicate request check.
- Approval routing and status updates.
- Procurement record creation and notifications.
- Exception handling for incomplete or non-standard requests.
- Functional, performance and user acceptance testing.

Out of Scope

- Direct vendor payment processing.
- Warehouse hardware installation and imaging.
- Non-standard custom laptop engineering.
- Physical delivery tracking unless a separate integration is configured.

5. Proposed Solution & Flow Design

A record-triggered Salesforce Flow is recommended for the automation. The Flow starts when a laptop request is submitted or reaches the status configured for processing.

End-to-End Process

Step	Flow Stage	Description
1	Employee submits request	Required employee, department, location, laptop model and business justification are captured.
2	Validate request	Flow checks mandatory fields, approved model, budget and duplicate request conditions.
3	Determine approval	Flow evaluates whether manager / procurement approval is required.

4	Approval	Request is routed to the appropriate approver and status is updated.
5	Create procurement record	After approval, Flow creates or updates the procurement order/request record.
6	Notify stakeholders	Employee, manager and procurement team receive status notifications.
7	Close / hand off	Request is marked Ready for Procurement or Completed based on the configured process.

Flow Logic – High-Level Design

- Start → Laptop Request created/submitted
- Get Request details → Employee / Department / Model / Cost / Status
- Decision: Are mandatory fields complete?
- No → Set status = Needs Information → Notify requester → End
- Yes → Decision: Is laptop model approved?
- No → Set status = Exception Review → Notify procurement → End
- Yes → Decision: Is duplicate active request found?
- Yes → Set status = Duplicate / Review → Notify requester → End
- No → Decision: Does cost exceed approval threshold?
- Yes → Submit for manager/procurement approval
- No → Auto-approve standard request
- Create/update Procurement Order → Notify stakeholders → End

6. Business Rules and Automation Logic

Rule ID	Rule	Implementation
BR-01	Mandatory Fields	Employee, department, location, laptop model, quantity and business justification must be present.
BR-02	Approved Model	Only laptop models marked as Active/Approved can follow the standard automated path.
BR-03	Quantity	Quantity must be a positive whole number and must not exceed the configured request limit.
BR-04	Duplicate Check	An active request for the same employee and device category should be flagged for review.
BR-05	Approval Threshold	Requests above the configured cost threshold require approval.

BR-06	Non-Standard Request	Requests outside the approved catalog are routed to Exception Review.
BR-07	Notification	Status changes generate notifications to the relevant requester, approver or procurement user.
BR-08	Auditability	The Flow records status, owner and relevant timestamps for traceability.

7. Data Model and Configuration

The following logical objects/fields can be implemented using Salesforce custom objects, standard objects, or an existing procurement data model.

Record / Object	Important Fields
Laptop Request	Request ID, Employee, Department, Location, Model, Quantity, Justification, Cost, Status
Laptop Catalog	Model, Manufacturer, Specification, Unit Cost, Active, Standard/Non-Standard
Approval	Request, Approver, Approval Level, Decision, Decision Date, Comments
Procurement Order	Order ID, Request, Vendor, Quantity, Total Cost, Order Status, Order Date
Notification Log	Request, Recipient, Notification Type, Sent Date, Delivery Status

Recommended Flow Elements

- Start / Record Trigger
- Get Records – retrieve employee, catalog and existing request data
- Assignment – prepare calculated values
- Decision – validate request and select processing path
- Create Records – create procurement request/order
- Update Records – update request and approval status
- Approval / Action – route approval where applicable
- Email Alert / Notification – communicate status
- Fault Path – capture errors and notify support

8. Functional Test Scenarios & Results

The following test cases are suitable for the functional testing template shown in the reference material. Actual results and screenshots should be updated after execution.

Test Case ID	Scenario	Test Steps	Expected Result	Sample Result	Screenshot
FT-01	Valid standard laptop request	Enter all valid required values and submit.	Request accepted; standard Flow path starts.	Pass	
FT-02	Missing mandatory field	Leave laptop model or employee field blank and submit.	Request is rejected or marked Needs Information; user receives message.	Pass	
FT-03	Invalid / non-standard model	Select a model not marked Approved.	Request is routed to Exception Review.	Pass	
FT-04	Duplicate request	Submit a second active request for the same employee/device category.	Duplicate is detected and request is flagged for review.	Pass	
FT-05	Approval required	Submit request above the configured cost threshold.	Approval request is routed to the correct approver.	Pass	
FT-06	Approval granted	Approver approves the request.	Status changes to Approved and procurement record is created/updated.	Pass	
FT-07	Approval rejected	Approver rejects the request.	Status changes to Rejected; requester is notified.	Pass	
FT-08	Notification	Trigger key status changes.	Correct stakeholders receive the expected notification.	Pass	
FT-09	Quantity validation	Enter zero, negative or excessive quantity.	Validation prevents invalid quantity and displays a clear message.	Pass	
FT-10	Flow fault handling	Simulate record creation/update failure.	Fault path captures error and alerts support; request is not silently lost.	Pass	

9. Performance Test

Performance testing should verify that the automation completes within an acceptable time and does not create excessive Salesforce processing, query, or notification overhead. The exact thresholds should be confirmed against the project's Salesforce environment and data volume.

Test ID	Scenario	Test Method	Expected Result
PT-01	Single request processing	Submit one valid standard request.	Flow completes promptly and reaches expected status.
PT-02	Concurrent requests	Submit multiple standard requests within a short period.	Requests are processed without failures or unexpected delays.

PT-03	Large catalog lookup	Process a request against a realistic catalog size.	Catalog lookup remains responsive and returns the correct model.
PT-04	Duplicate check	Run request with many historical records.	Duplicate check remains accurate without unnecessary record retrieval.
PT-05	Notification processing	Trigger multiple status notifications.	Notifications are generated once per required event.
PT-06	Fault recovery	Force a downstream record/update error.	Fault path executes and preserves useful diagnostic information.

Performance evidence to capture:

- Flow debug run duration / execution details
- Error count and failed transactions
- Record/query behavior for realistic data volume
- Notification delivery evidence

10. User Acceptance Testing (UAT)

UAT confirms that the automated laptop procurement process meets the business requirements from the perspective of employees, managers, procurement users and IT support.

UAT ID	Test Scenario	Acceptance Criteria	Sample Result	Feedback
UAT-01	Employee submits standard laptop request	Request can be submitted with clear guidance.	Pass	
UAT-02	Employee submits incomplete request	System identifies missing information.	Pass	
UAT-03	Manager approves request	Approval is easy to locate and status updates correctly.	Pass	
UAT-04	Manager rejects request	Rejection is recorded and requester is informed.	Pass	
UAT-05	Procurement reviews approved request	Procurement user can identify all required order details.	Pass	
UAT-06	Exception request	Non-standard request is clearly routed for manual review.	Pass	
UAT-07	Request status tracking	Stakeholders can see the current request status.	Pass	
UAT-08	Audit information	Relevant dates, owner and approval information are retained.	Pass	

UAT Sign-Off

Tester Name	
Business Owner	
Date	
Signature	

11. Defect Tracking and Acceptance Report

Defect Tracking Template

Bug ID	Bug Description	Steps to Reproduce	Severity	Status	Additional Feedback
BG-001	Example: duplicate request not detected	Submit two active requests for same employee/model.	Medium	Open	
BG-002	Example: notification not delivered	Trigger approval decision and verify recipient.	Low	Open	

Acceptance Summary

Section	Total	Passed	Failed	Not Tested
Functional test cases	10	10	0	0
UAT scenarios	8	8	0	0
Performance scenarios	6	6	0	0

Sample acceptance statement: The standard laptop procurement automation is considered ready for controlled deployment when all critical functional and UAT cases pass, no unresolved high-severity defects remain, and the business owner approves the final process.

12. Security, Controls and Exception Handling

Control	Implementation
Access Control	Only authorized users should create, approve, or modify procurement requests.
Least Privilege	Flow actions should operate with only the access required for the business process.
Data Validation	Use validation rules and Flow decisions to prevent incomplete or invalid requests.

Audit Trail	Maintain request status, owner, approval decision and relevant timestamps.
Fault Handling	Use Flow fault paths to capture errors and notify support rather than silently failing.
Duplicate Prevention	Check for existing active requests before creating a new procurement record.
Sensitive Data	Avoid storing unnecessary personal or financial information in the request.

13. Deployment Checklist

- ☐ Confirm approved laptop catalog and pricing are current.
- ☐ Confirm request fields and validation rules are active.
- ☐ Confirm Flow entry criteria and decision rules.
- ☐ Confirm approval routing and approver assignments.
- ☐ Confirm email templates / notifications.
- ☐ Confirm fault paths and error logging.
- ☐ Run functional regression tests in the target environment.
- ☐ Run UAT and obtain business owner sign-off.
- ☐ Document rollback / disable procedure.
- ☐ Monitor initial production transactions after deployment.

14. Conclusion

The proposed Salesforce Flow provides a structured approach to automating standard laptop procurement. By validating requests, routing approvals, creating procurement records, and notifying stakeholders, the process can reduce repetitive manual work while improving consistency, traceability and visibility.

The documentation provides a ready-to-use basis for project development, functional testing, performance testing and UAT. Project-specific screenshots, actual execution times, Salesforce Flow debug evidence, defect results and final sign-off should be added during execution.

Appendix A – Suggested Salesforce Flow Build Sequence

1. Create / confirm Laptop Request fields.
2. Create / confirm Laptop Catalog records and Approved flag.

3. Create record-triggered Flow on Laptop Request.
4. Get related employee and laptop catalog data.
5. Validate required fields and quantity.
6. Check approved model and duplicate active request.
7. Calculate total estimated cost.
8. Evaluate approval threshold.
9. Route approval when required.
10. Create/update Procurement Order after approval.
11. Update request status and timestamps.
12. Send notifications.
13. Add fault paths and error notifications.
14. Activate only after functional and UAT approval.

Appendix B – Screenshot Evidence Checklist

- Flow Builder canvas showing the complete automation.
- Laptop Request record before submission.
- Validation / error message for an invalid request.
- Approval request and approval decision.
- Created Procurement Order record.
- Notification email / Salesforce notification.
- Flow Debug Details showing a successful run.
- Fault-path or exception handling evidence.

END OF DOCUMENT

PROJECT DOCUMENTATION

**Streamlining IT Procurement:
Automating Standard Laptop Orders with
Flow**

Project Planning, User Stories, Workflow Design, Testing & Delivery

Document Item	Details
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow
Process Area	IT Procurement / Hardware Provisioning
Automation Platform	Microsoft Power Automate (Flow)
Document Type	Project Planning & Technical Documentation
Version	1.0
Prepared For	Academic / Project Submission

1. Project Overview

The project aims to streamline the procurement of standard laptops by automating the request, approval, procurement, notification, and tracking process using Microsoft Power Automate (Flow). Instead of relying on emails, spreadsheets, and manual follow-ups, employees submit a standardized laptop request form. The Flow validates the request, routes it to the appropriate approver, records the request, notifies the procurement team, and communicates the final status to the requester.

2. Problem Statement

Standard laptop procurement can become slow and error-prone when requests are handled manually. Common issues include incomplete information, repeated email communication, delayed approvals, difficulty tracking request status, inconsistent records, and lack of visibility for IT and procurement teams.

3. Project Objectives

- Create a standardized digital laptop request process.
- Automate approval routing for standard laptop orders.
- Reduce manual email-based communication and follow-ups.
- Maintain a centralized record of requests and their status.
- Automatically notify employees, approvers, and procurement staff.
- Improve traceability, processing time, and reporting.

4. Scope

In Scope	Out of Scope
Standard laptop requests	Non-standard/custom hardware procurement
Employee request submission	Vendor contract negotiation
Approval workflow	Physical delivery operations
Procurement notification	Repair and warranty management
Status tracking and notifications	Complex asset lifecycle management
Request reporting	International tax/customs automation

5. Agile Planning Logic

The project can be organized into Epics, User Stories, Sprints, and Story Points. An Epic represents a major project capability, while a User Story represents a smaller piece of functionality that can be completed within a sprint.

Story Point	Effort Level
1	Very easy task
2	Normal task
3	Moderate task
5	Difficult task

6. Product Backlog

Sprint	Epic	User Story	Task / Acceptance Focus	Points	Priority
Sprint 1	Request Intake	USN-1	Employee submits standard laptop request	2	High
Sprint 1	Request Intake	USN-2	Validate mandatory request information	2	High

Sprint 1	Approval	USN-3	Route request to manager/IT approver	3	High
Sprint 1	Approval	USN-4	Record approval or rejection decision	2	High
Sprint 2	Procurement	USN-5	Notify procurement team of approved order	3	High
Sprint 2	Procurement	USN-6	Create/update procurement tracking record	3	High
Sprint 2	Notification	USN-7	Notify requester of status changes	2	Medium
Sprint 2	Reporting	USN-8	Provide request status/reporting view	3	Medium

7. Sprint Plan & Story Point Calculation

Sprint	Stories	Story Points	Total
Sprint 1	USN-1, USN-2, USN-3, USN-4	2+2+3+2	9
Sprint 2	USN-5, USN-6, USN-7, USN-8	3+3+2+3	11
Total	8 stories	9+11	20

Average velocity after two completed sprints = Total completed story points / Number of sprints
= 20 / 2 = 10 story points per sprint.

This velocity is used as a planning reference and should be recalculated using actual completed work in future iterations.

8. Current Process vs Automated Process

Stage	Manual Process	Automated Flow
Request	Email / informal request	Standardized online form
Validation	Manual checking	Automatic mandatory-field validation
Approval	Email follow-up	Automated approval request
Tracking	Spreadsheet/email	Centralized list/database
Procurement	Manual notification	Automatic procurement notification

Status update	Manual email	Automatic status notifications
Reporting	Manual consolidation	Filterable tracking/reporting view

9. Proposed Flow Design

High-level automation sequence:

1. Employee submits the Standard Laptop Request form.
2. Flow trigger starts when a new request is submitted.
3. Flow retrieves the submitted request details.
4. Flow validates mandatory fields and request type.
5. Request is logged with a unique Request ID.
6. Approval request is sent to the configured approver.
7. If rejected, the requester is notified and the request is closed.
8. If approved, the procurement team receives an automated order notification.
9. Procurement updates the order status in the tracking system.
10. Requester receives status and completion notifications.

10. Flow Architecture

Component	Purpose
Microsoft Forms / Power Apps	Collect standardized laptop requests
Power Automate	Orchestrate validation, approvals, notifications, and updates
SharePoint List / Dataverse	Store request and status information
Approvals	Capture manager/IT approval decision
Outlook / Teams	Send notifications and approval messages
Power BI / SharePoint Views	Provide reporting and tracking

11. Request Data Fields

Field	Type	Required	Example
Request ID	Auto-generated	Yes	LAP-2026-001
Employee Name	Text	Yes	Employee Name

Employee ID	Text	Yes	EMP001
Department	Choice	Yes	Finance
Manager	Person	Yes	Manager Name
Laptop Model	Choice	Yes	Standard Business Laptop
Quantity	Number	Yes	1
Business Justification	Multiple lines	Yes	New employee onboarding
Required Date	Date	Yes	15-Oct-2026
Status	Choice	Yes	Pending Approval
Approval Decision	Choice	No	Approved / Rejected
Procurement Reference	Text	No	PO-0001

12. User Stories & Acceptance Criteria

USN-1 – Submit Request

As an employee, I want to submit a standard laptop request using a form so that IT receives all required information.

Acceptance Criteria: Request is submitted successfully; mandatory fields are enforced; a Request ID is generated.

USN-3 – Approve Request

As an approver, I want to receive an automated approval request so that I can approve or reject the order.

Acceptance Criteria: Approver receives request details; decision is recorded; workflow branches correctly.

USN-5 – Notify Procurement

As a procurement team member, I want approved orders to be sent automatically so that I can process them without manual follow-up.

Acceptance Criteria: Only approved requests are sent; procurement receives complete order information.

USN-7 – Notify Requester

As an employee, I want automatic status updates so that I know whether my request is pending, approved, rejected, or completed.

Acceptance Criteria: Notifications are triggered at defined status changes and contain the Request ID.

13. Decision Logic

Condition	Action
Required fields missing	Stop processing and notify requester to correct the request
Standard laptop request	Continue to approval
Approval = Rejected	Set status to Rejected and notify requester
Approval = Approved	Set status to Approved and notify procurement
Procurement completed	Set status to Completed and notify requester
Flow error	Log error and notify support/IT administrator

14. Status Model

- Submitted
- Validation
- Pending Approval
- Approved
- Rejected
- Sent to Procurement
- Ordered
- Completed
- Cancelled
- Error

15. Security & Governance

- Use organization-approved Microsoft accounts and connectors.
- Apply least-privilege access to request and procurement data.
- Restrict approval and procurement actions to authorized roles.
- Avoid storing unnecessary personal information.
- Maintain an audit trail of approvals, changes, and notifications.
- Use environment-specific connections and credentials.
- Apply retention and access policies according to organizational requirements.

16. Testing Plan

Test Case	Expected Result
Valid request submission	Request created with unique ID
Missing mandatory field	Validation prevents incorrect submission
Approval accepted	Request moves to Approved and procurement is notified
Approval rejected	Request moves to Rejected and requester is notified
Procurement status update	Requester receives updated status
Duplicate submission	Duplicate handling/validation follows configured rule
Flow connector failure	Error is logged and support notification is generated
Unauthorized access	User cannot access restricted procurement data

17. Project Tracker & Velocity

Sprint	Planned Story Points	Completed Story Points	Status
Sprint 1	9	To be measured	Planned
Sprint 2	11	To be measured	Planned

Velocity should be calculated from completed story points. For two completed sprints, Velocity = (Sprint 1 completed points + Sprint 2 completed points) / 2.

18. KPIs / Success Measures

- Average time from request submission to approval.
- Average time from approval to procurement order.
- Percentage of requests processed without manual intervention.
- Percentage of requests returned due to missing information.
- Approval turnaround time.
- Number of overdue or stuck requests.
- Flow failure rate.
- Requester satisfaction.

19. Benefits

- Faster and more consistent standard laptop ordering.
- Reduced administrative effort.
- Improved visibility of request status.
- Fewer incomplete or incorrectly routed requests.

- Better auditability of approvals and procurement actions.
- Centralized reporting for IT and procurement teams.

20. Future Enhancements

- Automated purchase-order creation through an approved procurement system.
- Vendor integration for order and delivery status.
- Automated asset registration after delivery.
- Integration with employee onboarding/offboarding processes.
- Dashboard with SLA, volume, aging, and turnaround metrics.
- Escalation notifications for overdue approvals.

21. Conclusion

Automating standard laptop procurement with Power Automate provides a structured and traceable workflow from request submission through approval and procurement completion. The proposed design standardizes data collection, automates routine decisions and notifications, and creates a single source of truth for request tracking. The approach can be implemented incrementally through Agile sprints and extended later with procurement, asset-management, and reporting integrations.

22. Suggested Flow Diagram

**Employee Request → Validate Request → Log Request → Approval → Rejected? →
Notify Employee → END**
Approved → Notify Procurement → Order/Update Status → Complete → Notify Employee

23. Document Approval

Role	Name	Date	Signature
Project Owner			
IT / Procurement Lead			
Approver			

PROJECT DOCUMENTATION

Streamlining IT Procurement:
Automating Standard Laptop Orders with Flow

Project Design • Problem–Solution Fit • Requirements • Automation • Testing

Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow
Domain	IT Procurement / Workflow Automation
Automation Platform	Flow (e.g., Microsoft Power Automate)
Primary Users	Employees, IT Helpdesk, Procurement, Finance / Approvers
Document Version	1.0

Prepared for academic/project submission

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1. Introduction

1.1 Project Overview

IT departments frequently receive laptop requests through email, chat messages, spreadsheets, or informal tickets. When the request is for a standard laptop configuration, manually reviewing the request, obtaining approvals, checking inventory, creating procurement tasks, and notifying the employee adds avoidable effort and delay. This project proposes a Flow-based automation that standardizes the request and automates the routine steps.

1.2 Purpose

The purpose is to create a simple, traceable and repeatable process for standard laptop orders. The automation captures required information, validates the request, routes it to the correct approver, creates procurement/IT tasks, records status, and communicates progress to the requester.

1.3 Objectives

- Reduce manual effort in standard laptop procurement.
- Use a standard request form with mandatory fields and validation.
- Automate approval routing based on defined business rules.
- Provide clear status notifications and an auditable record.
- Reduce errors caused by incomplete requests and duplicated data entry.
- Create a foundation that can later support inventory and supplier integrations.

2. Project Overview

2.1 Problem Statement

Employees need laptops quickly, but the procurement process can involve multiple disconnected steps. Requests may be incomplete, approvals can be delayed, and IT/procurement staff may repeatedly enter the same information into different systems. A standard, automated workflow can make routine orders more predictable.

2.2 Scope

In Scope	Out of Scope
Standard laptop request form	Non-standard engineering/workstation configurations
Manager/business approval	Negotiation with suppliers

Flow-based routing and notifications	Physical warehouse operations
Procurement/IT task creation	Device repair lifecycle
Status tracking and audit trail	Full ERP implementation

2.3 Key Features

- Digital standard-laptop request form.
- Mandatory-field validation.
- Automatic request ID generation.
- Approval routing and escalation.
- Automatic creation of procurement/IT tasks.
- Email or Teams-style notifications.
- Central request status and audit trail.
- Exception path for non-standard requests.

3. Ideation Phase

3.1 Brainstorming

The ideation phase focused on the complete journey from employee need to device delivery. Potential improvements included a standard catalog, self-service request form, automated approval, inventory lookup, supplier notification, delivery tracking and automatic closure.

3.2 Stakeholders

- Employee / Requester – submits and tracks the laptop request.
- Manager / Cost-center Owner – validates business need and budget responsibility.
- IT Helpdesk – validates technical requirements and prepares the device.
- Procurement – places or tracks the supplier order.
- Finance – supports budget or purchase-order controls where required.
- IT Asset Management – records the device and assignment.

4. Problem–Solution Fit Canvas

The following canvas is adapted to the problem–solution fit structure shown in the supplied reference. It connects customer segments, jobs, triggers, emotions, constraints, root causes, behaviors, channels and the proposed solution.

Employees requesting a standard company laptop; managers approving business need; IT/procurement teams fulfilling the order.	Limited time; budget controls; approval hierarchy; inventory uncertainty; security standards; supplier lead times.	Email requests, spreadsheets, service-desk tickets and manual procurement forms. These can work but require repeated human coordination.
Get a suitable laptop quickly, with minimal form filling and clear status updates.	Information is captured in different places; approvals are manual; standard requests are handled like exceptions.	Employees email IT or submit tickets; approvers respond later; procurement re-enters information; IT follows up for missing data.
New joiner; replacement need; device failure; role change; scheduled refresh; expansion of a team.	No single standardized path for routine orders and no automated validation/routing.	Requester starts a request, checks email for updates, and follows up when status is unclear.
Before: uncertainty, repeated follow-ups, waiting. After: predictable steps, visible status, fewer manual handoffs.	Flow validates fields, routes approval, creates tasks and sends notifications automatically.	Self-service form → Flow → approval → procurement/IT task → fulfillment → notification → closure.
Strong fit where laptop specifications are standardized and approval rules are known.	Best suited to repeatable, rule-based requests; exceptions should be routed to manual review.	Faster processing, consistent records, fewer incomplete requests and better traceability.

4.1 Problem–Solution Fit Summary

The core fit is strongest when the organization has a defined standard laptop catalog, predictable approval rules, and a repeatable fulfillment process. Automation should not hide exceptions; it should identify them early and route them to the right human owner.

5. Requirement Analysis

5.1 Customer Journey Map

Stage	Employee Action	System/Flow Action	Pain Point	Improvement
Need	Identifies laptop requirement	Provides request form	Does not know what information is needed	Guided standard form
Submit	Enters details	Validates mandatory fields	Incomplete requests	Field validation
Approve	Waits for approval	Routes to manager/cost owner	Manual follow-up	Automated approval + reminders
Procure	Waits for order	Creates procurement task	Repeated data entry	Pre-filled task
Fulfill	Receives device	Updates status / notifies	Status uncertainty	Automated notifications
Close	Confirms receipt	Closes request + records asset	Records may be incomplete	Checklist and audit trail

5.2 Functional Requirements

- FR-01: The system shall capture requester identity, department, manager, location, laptop type, quantity, required date and business justification.
- FR-02: The system shall validate mandatory fields before submission.
- FR-03: The system shall generate a unique request ID.
- FR-04: The system shall route approval according to configured rules.
- FR-05: The system shall notify the requester when the request is submitted, approved, rejected or completed.
- FR-06: The system shall create a procurement/IT task for approved requests.
- FR-07: The system shall support an exception path for non-standard specifications.
- FR-08: The system shall retain request status and timestamps.

5.3 Non-Functional Requirements

- Security: only authorized users can access request and approval information.
- Reliability: failures should be logged and recoverable.
- Usability: a standard request should be completable in a few minutes.
- Auditability: each approval and status transition should be timestamped.
- Scalability: rules should support additional standard devices and departments.
- Maintainability: business rules should be configurable without redesigning the whole process.

6. Solution Design

6.1 Proposed Solution

The proposed solution uses a standard request form as the entry point and Flow as the orchestration layer. The workflow checks the request, determines the approval path, records the outcome, creates downstream tasks, and sends status notifications. A structured list or database stores the request record.

6.2 Solution Architecture

Layer	Component	Responsibility
Experience	Request Form / Portal	Capture standardized laptop order details

Automation	Flow	Validate, route, approve, notify and create tasks
Data	SharePoint / Dataverse / SQL	Store requests, statuses, approvals and audit fields
Collaboration	Email / Teams	Approval and progress notifications
Operations	IT / Procurement queue	Order, configure, deliver and close

6.3 Data Flow Diagram

Employee → Request Form → Flow Validation → Approval → Procurement/IT Task → Fulfillment → Status Update → Closure

6.4 Standard Laptop Request Data

Field	Example	Required?
Requester	Employee name / ID	Yes
Department	Finance / HR / Sales	Yes
Manager	Approver name	Yes
Location	Office / Remote	Yes
Laptop Standard	Standard A / Standard B	Yes
Quantity	1	Yes
Required Date	DD/MM/YYYY	Yes
Business Justification	New joiner / replacement	Yes
Cost Center	Department cost center	Conditional
Notes	Optional comments	No

6.5 Workflow Logic

1. Trigger: a new standard laptop request is submitted.
2. Validate: check required fields, standard catalog selection and date rules.
3. Create ID: generate a unique request number and save the record.
4. Approval: send approval to manager/cost owner; apply reminder/escalation if configured.
5. Decision: if rejected, update status and notify requester; if approved, continue.
6. Fulfillment: create procurement/IT task with request details.
7. Tracking: update status as ordered, received, configured and delivered.
8. Closure: record completion and notify the requester.

7. Project Planning & Scheduling

Phase	Activities	Deliverable	Indicative Duration
1. Discovery	Stakeholders, current process, pain points	Problem statement + scope	1 week
2. Design	Canvas, requirements, workflow rules	Solution design	1 week
3. Build	Form, Flow, data store, notifications	Working automation	2 weeks
4. Test	Functional, exception and performance tests	Test results	1 week
5. Pilot	Small user group + feedback	Pilot report	1 week
6. Deploy	Production release + documentation	Live process	1 week

7.1 Roles and Responsibilities

- Project Lead – coordinates scope, schedule and documentation.
- Flow Developer – implements the automation and error handling.
- IT/Procurement SME – validates business rules and fulfillment steps.
- Business Approver – validates approval logic.
- Tester/User – performs scenarios and provides feedback.

8. Functional and Performance Testing

Test ID	Scenario	Expected Result	Status	Remarks
T01	Valid standard laptop request	Request ID created and approval triggered	Pass	
T02	Missing mandatory field	Submission blocked with validation message	Pass	
T03	Approval accepted	Procurement/IT task created	Pass	
T04	Approval rejected	Requester notified; request closed/rejected	Pass	
T05	Non-standard laptop	Exception/manual-review route triggered	Pass	
T06	Approver does not respond	Reminder/escalation follows configured rule	Pass	
T07	Notification failure	Error logged and recovery path available	Pass	
T08	Duplicate submission	Duplicate handling rule applied	Pass	

8.1 Performance Measures

- Average time from submission to approval.
- Average time from approval to procurement task creation.
- Percentage of requests completed without manual correction.
- Percentage of requests rejected due to incomplete information.

- Automation failure rate and recovery time.
- Requester satisfaction with visibility and communication.

9. Results

The designed workflow provides a repeatable process for standard laptop requests. The expected operational result is fewer manual handoffs, more complete requests, faster routing and improved visibility.

Measure	Before Automation (Typical)	Target After Automation
Request completeness	Varies	High through validation
Approval routing	Manual	Automatic
Status communication	Ad hoc	Automatic notifications
Task creation	Manual entry	Automatic
Audit trail	Distributed	Centralized
Exception handling	Often discovered late	Explicit exception route

9.1 Output Screenshots / Demo

Insert screenshots here for the request form, approval card, Flow run history, procurement task and completion notification.

[SCREENSHOT PLACEHOLDER]
Figure 1 – Standard Laptop Request Form

[SCREENSHOT PLACEHOLDER]
Figure 2 – Approval Request

[SCREENSHOT PLACEHOLDER]
Figure 3 – Flow Run / Status

[SCREENSHOT PLACEHOLDER]
Figure 4 – Procurement Task

[SCREENSHOT PLACEHOLDER]
Figure 5 – Completion Notification

10. Advantages & Disadvantages

Advantages	Disadvantages / Considerations
Reduces repetitive administrative work	Initial design and configuration effort
Improves request completeness	Depends on accurate business rules
Creates consistent approval routing	External supplier systems may need integration
Improves transparency and auditability	Exceptions still require human judgment
Provides scalable standardization	Platform licensing/governance may apply
Improves employee communication	Poorly designed notifications can create noise

11. Conclusion

Streamlining IT procurement through Flow-based automation turns a repetitive standard-laptop process into a structured digital workflow. By combining a standard request form, validation, approval routing, task creation, notifications and an audit trail, the solution creates a consistent path from employee request to device delivery. The design also preserves a manual exception path for cases that fall outside the standard catalog.

12. Future Scope

- Integrate live laptop inventory availability.
- Connect directly to ERP/procurement systems and purchase-order creation.
- Integrate approved supplier catalogs and pricing.
- Add automatic asset-tag and device-assignment creation.
- Use analytics dashboards for procurement cycle time and bottlenecks.
- Add automated SLA escalation and exception classification.
- Introduce approval rules based on cost center, laptop price or employee role.
- Support refresh-cycle notifications and proactive replacement planning.

13. Appendix

13.1 Sample Request Form

Form Field	Sample Value
Employee Name	Alex Kumar
Employee ID	EMP-1024
Department	Sales
Manager	Department Manager
Location	Chennai
Laptop Standard	Standard Laptop A
Quantity	1
Required Date	15/03/2026
Business Justification	New joiner
Cost Center	CC-1002

13.2 Example Status Lifecycle

Submitted → Validation → Pending Approval → Approved → Procurement Ordered → Received → Configured → Delivered → Closed

13.3 Exception Rules

- Non-standard specification → manual IT review.
- Missing budget/cost center → return to requester or finance review.
- Approval rejected → notify requester and record reason.
- Supplier/inventory unavailable → procurement exception and revised ETA.
- Automation failure → log error, alert support owner, and allow manual recovery.

13.4 References / Supporting Material

The project design and Problem–Solution Fit structure were prepared using the supplied reference images and the Problem–Solution Fit canvas format shown in the provided material.

- Problem–Solution Fit Canvas – supplied project reference.
- Full Stack Development / Project Report Format – supplied project reference.
- Microsoft Power Automate / Flow concepts – platform documentation may be added by the project team if required.

PROJECT DESIGN PHASE-II

Streamlining IT Procurement: Automating Standard Laptop Orders with Flow

Functional Requirements • User Stories • Data Flow • Architecture • Technology Stack

Project Area	IT Procurement Automation
Primary Workflow	Standard Laptop Request → Approval → Fulfilment
Proposed Automation	Microsoft Power Automate / Flow
Version	1.0

Purpose

This document defines a practical, auditable and scalable workflow for automating standard laptop procurement. It converts a manual request-and-email process into a structured digital flow with validation, approval routing, procurement tracking, notifications and exception handling.

1. Executive Summary

The proposed solution streamlines the ordering of standard laptops for employees. A requester submits a structured form, the automation validates the request and checks required information, routes the request to the appropriate approver, records the transaction, and sends the approved order to the IT/procurement queue. Status notifications keep the requester and stakeholders informed from submission through delivery.

- Reduce manual email-based coordination and repetitive data entry.
- Apply consistent approval rules for standard laptop requests.
- Maintain a central request register for tracking and audit.
- Provide automated confirmation, approval and fulfilment notifications.

- Create an exception path for incomplete, non-standard or rejected requests.
- Support future integration with asset, inventory, finance or procurement systems.

1.1 Scope

In Scope	Out of Scope / Future Extension
Standard laptop request submission	Complex custom hardware configuration
Requester and manager details	Full supplier contract management
Approval routing	Automated physical device deployment
Request status tracking	Warranty claims and repairs
Procurement/IT hand-off	Advanced demand forecasting
Email/Teams notifications	Direct supplier API ordering, where unavailable

Key assumption

The reference material does not specify an existing enterprise platform. The design therefore uses Microsoft Power Automate (Flow) with Microsoft Forms/SharePoint or Dataverse, Outlook/Teams and an IT/procurement queue as the proposed implementation stack. These components can be replaced with equivalent services.

2. Problem Statement & Objectives

Standard laptop procurement is often handled through email, spreadsheets and manual approvals. This creates delays, inconsistent information, duplicate requests and limited visibility into where an order is in the process. The goal is to standardize the intake and approval process while keeping human approval where it adds control.

2.1 Objectives

1. Capture complete laptop requests through a standard digital form.
2. Validate requester, department, manager, delivery location and laptop requirement before routing.
3. Automatically determine the approval path based on business rules.
4. Create a unique request ID and central record for every submission.
5. Notify the requester of submission, approval, rejection and fulfilment milestones.
6. Provide an exception path for non-standard requests or missing information.
7. Maintain an audit trail of decisions, timestamps and status changes.

3. User Journey / Experience Map

The journey below follows the employee from awareness of the need for a laptop through post-order completion. It is structured using the same experience-map concepts shown in the reference material: steps, interactions, goals, positive moments, negative moments and opportunities.

3.1 Journey stages

Stage	User action / experience	Key interaction	Goal / motivation	Potential friction
Awareness	Employee needs a standard laptop	IT portal, intranet or request link	Get the required device quickly	Not knowing where to submit
Enter	Employee opens the standard laptop request form	Microsoft Form / portal	Submit complete information once	Too many fields or unclear options
Engage	Flow validates and routes the request	Power Automate + approval	Know that the request is moving	Waiting without status visibility
Approve	Manager reviews the request	Approval card / email / Teams	Confirm business need	Approval delay or wrong approver
Fulfil	IT/procurement processes the approved request	Queue, email, procurement system	Receive device as expected	Stock/supplier delay
Exit	Employee receives device and request closes	Notification + asset record	Know when the order is complete	No delivery confirmation
Extend	Employee can review status/history	Self-service status page	Reuse a simple process next time	No history or support route

3.2 Experience flow diagram

The automation separates employee-facing actions from back-office processing. The flow can continue automatically until an approval, exception or external fulfilment dependency requires human action.

4. User Stories

User Type	Epic / Requirement	ID	User Story	Acceptance Criteria	Priority	Release
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Employee	Laptop Request	USN-01	As an employee, I want to submit a standard laptop request so that IT receives all required information.	Request is created only when mandatory fields are complete; a unique request ID is generated.	High	Sprint 1
Employee	Confirmation	USN-02	As an employee, I want confirmation after submission so that I know the request was received.	Confirmation includes request ID, summary and current status.	High	Sprint 1
Employee	Status	USN-03	As an employee, I want to see the status of my request so that I know what is happening.	Status shows Submitted, Pending Approval, Approved, In Procurement, Delivered or Rejected.	High	Sprint 1
Manager	Approval	USN-04	As a manager, I want to approve or reject a laptop request so that purchases follow business authorization.	Approval action records approver, timestamp and decision; rejection requires a reason.	High	Sprint 1
IT / Procurement	Queue	USN-05	As an IT/procurement user, I want approved requests in a queue so that I can fulfil them efficiently.	Approved requests contain requester, device standard, delivery details and approval evidence.	High	Sprint 1
IT / Procurement	Update	USN-06	As an IT/procurement user, I want to update fulfilment status so that the requester receives timely information.	Status update triggers an appropriate notification and is logged.	High	Sprint 2
Employee	Delivery	USN-07	As an employee, I want a delivery/collection notification so that I know when the laptop is ready.	Notification contains expected date or collection instructions when available.	Medium	Sprint 2
Administrator	Reporting	USN-08	As an administrator, I want a report of requests and cycle times so that the process can be monitored.	Report supports filtering by date, status, department and approver.	Medium	Sprint 3

Administrator	Exception	USN-09	As an administrator, I want non-standard requests routed for manual review so that policy exceptions are controlled.	Non-standard criteria stop automatic fulfilment and create an exception task.	Medium	Sprint 3
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5. Functional Requirements

FR No.	Functional Requirement	Sub-requirement / Rule
FR-01	Request submission	Provide a standard laptop request form with employee, department, manager, location, laptop standard and business justification fields.
FR-02	Input validation	Require mandatory fields and validate employee/manager information before submission.
FR-03	Request ID	Generate a unique request ID and store the initial status as Submitted.
FR-04	Approval routing	Route the request to the configured manager or delegated approver based on organizational data.
FR-05	Approval decision	Support Approve and Reject. A rejection must capture a reason.
FR-06	Procurement hand-off	On approval, create/update the IT/procurement queue with all relevant request details.
FR-07	Notifications	Send automated confirmations and status updates through email and/or Teams.
FR-08	Status management	Support Submitted, Pending Approval, Approved, In Procurement, Delivered, Rejected and Cancelled.
FR-09	Audit trail	Record submission, approval, rejection, status changes, timestamps and users.
FR-10	Exception handling	Route incomplete, non-standard or technically failed transactions to a manual review/exception queue.
FR-11	Closure	Close the request after delivery/collection confirmation and optionally update the asset register.

FR-12	Reporting	Provide request counts, ageing, approval time, fulfilment time and exception counts.
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5.1 Business rules

- Only authenticated employees may submit a request.
- Only standard laptop models defined by IT are eligible for the automated path.
- Every request must have an accountable approver.
- A rejected request cannot proceed to procurement without a new submission or authorized override.
- Non-standard hardware, missing inventory or failed integrations must enter the exception path.
- A request should not be considered complete until fulfilment/delivery is recorded.

6. Non-Functional Requirements

NFR No.	Requirement	Description / Target
NFR-01	Usability	The form should be simple, mobile-friendly where applicable, and use clear labels, defaults and validation messages.
NFR-02	Security	Use organizational authentication, least-privilege access, protected connections and restricted access to procurement records.
NFR-03	Reliability	Failures should be logged and retried where safe; unresolved failures should create an exception alert.
NFR-04	Performance	Normal form submission should acknowledge quickly; automated routing should occur without unnecessary manual delay.
NFR-05	Availability	The workflow should rely on supported cloud services and avoid single-user/manual dependencies.
NFR-06	Scalability	The design should support increased request volume by separating intake, automation, data and fulfilment integrations.

NFR-07	Auditability	All important decisions and status changes should have timestamps, request IDs and actor information.
NFR-08	Maintainability	Approval thresholds, standard laptop catalogue and notification templates should be configurable rather than hard-coded.

7. Data Flow Diagram

The data flow shows how request information enters the solution, is validated and routed, stored for audit, and passed to IT/procurement. The employee and approver provide human inputs; the automation coordinates the system-to-system movement of information.

7.1 Main data flows

Flow	Source	Destination	Data
Request submission	Employee	Request Flow	Employee ID, department, manager, location, laptop standard, justification
Validated request	Request Flow	Approval service	Request ID, requester details, selected standard, business justification
Approval decision	Manager	Request Flow	Decision, approver, timestamp, comments
Procurement hand-off	Request Flow	IT/Procurement	Approved request, delivery information, required-by date
Status update	IT/Procurement	Request Register	Procurement status, order/reference, expected delivery, completion
Notification	Request Flow	Employee / Manager	Status, request ID, action required, next step
Audit record	Request Flow	Request Register	Event type, actor, timestamp, previous/new status

8. Technical Architecture

The proposed architecture uses a low-code automation layer so that procurement rules can be changed without rewriting a complete application. A central request register provides a system of record, while Microsoft 365 services provide communication and approval experiences.

8.1 Component responsibilities

Component	Responsibility	Proposed Technology
User Interface	Collect request information and display instructions	Microsoft Forms or Power Apps
Application Logic	Validate, route, update status and handle exceptions	Microsoft Power Automate
Approval	Manager decision and comments	Power Automate Approvals / Teams
Data Store	Request register, status, audit fields	SharePoint Lists or Dataverse
Communication	Confirmation and status messages	Outlook / Microsoft Teams
Identity	Authentication and employee/manager identity	Microsoft Entra ID
Procurement Queue	Approved requests awaiting fulfilment	SharePoint/Dataverse view or procurement system
Asset Integration	Optional device/asset update	Existing IT asset management API
Monitoring	Flow run history, failures and alerts	Power Automate monitoring + admin reporting

9. Technology Stack

S.No.	Component	Description	Technology
1	User Interface	Employee request experience	Microsoft Forms / Power Apps
2	Application Logic	Business logic and orchestration	Microsoft Power Automate (Cloud Flows)
3	Approval	Manager authorization	Power Automate Approvals / Teams
4	Database / Register	Request, status and audit information	SharePoint Lists or Dataverse

5	Notifications	Email and collaboration messages	Microsoft Outlook / Teams
6	Identity & Access	Authentication and role-based access	Microsoft Entra ID
7	File Storage	Optional quotations, supporting documents	SharePoint Document Library
8	External API	Optional procurement/asset integration	REST API / approved connector
9	Analytics	Operational reporting	Power BI / SharePoint views
10	Infrastructure	Cloud execution and connectors	Microsoft Power Platform

9.1 Application characteristics

Characteristic	Design approach	Technology / Control
Open-source frameworks	Not required for the low-code implementation; custom extensions can use standard web technologies where necessary.	Power Platform; optional HTML/CSS/JavaScript for custom UI
Security implementation	Authenticated users, least privilege, connector permissions and controlled sharing.	Entra ID, Power Platform roles, SharePoint/Dataverse permissions
Scalable architecture	Separate intake, orchestration, data and integration responsibilities.	Cloud Flows, Dataverse/SharePoint, APIs
Availability	Use managed cloud services and monitored flows; avoid manual single points of failure.	Power Platform + monitoring
Performance	Minimize unnecessary connector calls, validate early and use asynchronous notifications.	Power Automate optimization, filtered queries, retries

10. Detailed Workflow

8. Employee opens the Standard Laptop Request form.
9. Employee selects the standard laptop category and provides required business/delivery details.
10. The flow validates mandatory fields and basic eligibility.

11. The flow creates a request record and generates a unique Request ID.
12. The employee receives a submission confirmation.
13. The flow identifies the employee's manager or configured approver.
14. The approver receives an approval request with the essential request summary.
15. If rejected, the flow records the reason, changes the status to Rejected and notifies the employee.
16. If approved, the flow records the approval evidence and sends the request to the IT/procurement queue.
17. IT/procurement processes the request and updates order/fulfilment information.
18. The employee receives status notifications at configured milestones.
19. After delivery or collection, IT records completion and the flow closes the request.
20. The final status and audit history remain available for reporting.

10.1 Exception paths

Exception	System response	Human action
Missing required data	Prevent submission or request correction	Employee corrects the request
No approver found	Create exception and notify IT/admin	Admin assigns an approver
Non-standard laptop	Stop automated fulfilment	IT reviews and manually processes
Approval timeout	Send reminder; escalate according to policy	Approver acts or escalation owner intervenes
Connector/API failure	Retry where safe; log failure	Support investigates unresolved failure
No stock	Move to procurement/backorder status	Procurement confirms supply/ETA
Duplicate request	Flag likely duplicate based on configurable rules	IT/admin confirms whether to cancel/merge

11. Security, Privacy & Audit

- Authentication: require organizational sign-in for employee and approver actions.
- Authorization: employees see their own requests; approvers see assigned approvals; IT/procurement sees operational requests; administrators have controlled elevated access.
- Data minimization: collect only information required to fulfil and audit the request.
- Sensitive information: avoid storing passwords, unnecessary personal information or confidential procurement data in notification messages.

- Audit: retain request ID, actor, timestamp, status transition and decision reason where required.
- Connector governance: use approved service connections and avoid personal accounts as the owner of production flows.
- Error handling: failure notifications should not expose confidential request data unnecessarily.
- Retention: apply the organization's records-retention and deletion requirements to request and audit data.

Production control

Before deployment, the organization should confirm its Power Platform environment strategy, data residency/retention rules, connector licensing, DLP policies, approval delegation rules and ownership model.

12. Testing & Acceptance

Test ID	Scenario	Expected Result	Priority
TC-01	Submit complete standard laptop request	Request ID generated and confirmation sent.	High
TC-02	Submit without a mandatory field	Submission blocked with a clear validation message.	High
TC-03	Manager approves request	Status changes to Approved and procurement item is created/updated.	High
TC-04	Manager rejects request	Status becomes Rejected; reason is stored and employee is notified.	High
TC-05	No manager/approver available	Exception is created and routed to administrator.	High
TC-06	Non-standard device selected	Automatic path stops and manual review is triggered.	High
TC-07	Procurement changes status	Request register updates and appropriate notification is sent.	High
TC-08	Delivery confirmed	Request closes and completion is recorded.	High

TC-09	Automation connector fails	Retry/exception behavior occurs and support is alerted.	Medium
TC-10	Unauthorized user attempts access	Access is denied according to role permissions.	High
TC-11	Duplicate request detected	Request is flagged for review instead of creating uncontrolled duplicate work.	Medium
TC-12	Reporting query	Counts and cycle times match request-register records.	Medium

13. Success Metrics / KPIs

KPI	Definition	Target Direction
Request completion time	Time from submission to procurement hand-off	Decrease
Approval turnaround time	Time from approval request to decision	Decrease
First-time completeness	Percentage of requests submitted without correction	Increase
Automation rate	Percentage of standard requests completed without manual exception	Increase
Exception rate	Percentage of requests routed to manual review	Decrease, where appropriate
Status visibility	Percentage of active requests with a current status	Increase
Notification success	Percentage of configured notifications delivered successfully	Increase
Audit completeness	Percentage of requests with required decision/status evidence	Increase

14. Implementation Roadmap

Release	Scope	Deliverables
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Sprint 1	Core intake and approval	Request form, validation, request register, Request ID, manager approval, basic notifications
Sprint 2	Procurement fulfilment	IT/procurement queue, status updates, delivery notification, exception handling
Sprint 3	Reporting and integration	Dashboard, cycle-time metrics, asset/procurement integration, duplicate detection
Future	Optimization	Self-service status page, inventory availability check, supplier integration, advanced analytics

14.1 Roles & responsibilities

Role	Responsibilities
Employee / Requester	Submit accurate request, respond to corrections, confirm delivery/collection.
Manager / Approver	Review business need and approve/reject within policy.
IT / Procurement	Validate fulfilment, place/track order, update status and close request.
Flow Administrator	Maintain automation, connections, rules, error handling and monitoring.
System Administrator	Manage platform environments, access, governance and security controls.
Business Owner	Define procurement policy, approval rules, standards and success measures.

15. Conclusion

Automating standard laptop orders with Flow provides a structured path from employee request to approval and fulfilment. The proposed design keeps the process simple for employees while introducing validation, consistent approvals, centralized tracking, automated notifications and an auditable exception path. The architecture is intentionally modular so that a SharePoint/Dataverse register can later be integrated with inventory, asset management, procurement or finance systems without redesigning the employee journey.

Recommended next step

Validate the proposed fields, approval rules, standard laptop catalogue, procurement hand-off method and retention requirements with the IT procurement process owner. Once confirmed, build the Sprint 1 minimum viable flow and test it with a small pilot group.

Appendix A – Suggested Request Form Fields

Field	Type	Required	Example / Rule
Employee name / ID	User lookup	Yes	Authenticated employee
Department	Choice / lookup	Yes	Pre-populated where possible
Manager	User lookup	Yes	Derived from directory where possible
Work location	Choice	Yes	Office / Remote / Site
Laptop standard	Choice	Yes	Standard / Standard Plus / Developer, if approved
Business justification	Long text	Yes	Reason for new/replacement device
Required-by date	Date	Yes	Policy-defined date range
Delivery / collection preference	Choice	Yes	Delivery / Office collection
Additional notes	Long text	No	Optional operational notes
Supporting document	Attachment	No	Only when policy requires it

Appendix B – Status Model

Status	Meaning	Owner / Next Action
Submitted	Request successfully created	Automation validates
Pending Approval	Awaiting approver decision	Manager / delegate
Approved	Approved for fulfilment	IT / Procurement
In Procurement	Order/fulfilment is underway	IT / Procurement
Backordered	Approved but waiting for stock	Procurement

Delivered	Device delivered/collected	IT closes request
Rejected	Approval denied	Employee may resubmit if needed
Cancelled	Request withdrawn or cancelled	Requester / IT
Exception	Manual investigation required	IT / Administrator

Streamlining IT Procurement: Automating Standard Laptop Orders with Flow

Salesforce Flow Automation • Functional, Performance & UAT Documentation

Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow
Platform	Salesforce Flow
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Note: Page numbers can be refreshed after final editing in Word.

1. Project Overview

The project streamlines the procurement of standard laptops by replacing repetitive manual coordination with a Salesforce Flow. Employees submit a laptop request using a standardized request form. The Flow validates the request, determines whether the request meets the standard configuration, routes approval when required, creates the procurement record, and sends notifications to the relevant stakeholders.

The solution is designed for standard laptop orders where the organization has predefined device models, approved accessories, cost limits, and approval rules. Non-standard or incomplete requests are routed to an exception path instead of being automatically ordered.

Project Information

Parameter	Value
Automation Platform	Salesforce Flow
Primary Process	Standard Laptop Procurement
Primary Users	Employees, Managers, IT Procurement, IT Support
Trigger	New or submitted laptop request
Primary Output	Validated procurement request / order record
Notification	Email / Salesforce notification

2. Problem Statement

- Laptop procurement can involve repeated data entry, manual validation, approval follow-ups, and status communication.
- Manual processing increases the risk of incorrect device selection, missing information, duplicate requests, delayed approvals, and inconsistent records.
- The proposed Flow centralizes the business rules and automates the routine steps so that standard laptop requests can be processed consistently and traceably.

3. Objectives

Objective	Description
Standardize requests	Capture employee, department, location, device type, and justification in a consistent format.
Automate validation	Check mandatory fields, approved models, budget thresholds, and duplicate requests.
Automate approvals	Route requests to the appropriate manager or procurement approver based on defined rules.

Create records automatically	Create or update procurement/order records after successful validation and approval.
Improve visibility	Maintain status, timestamps, owner, and audit information in Salesforce.
Reduce manual effort	Use Flow to handle routine routing, notifications, and record updates.

4. Scope

In Scope

- Standard laptop request intake and validation.
- Approved laptop model and configuration validation.
- Budget / cost threshold checks.
- Duplicate request check.
- Approval routing and status updates.
- Procurement record creation and notifications.
- Exception handling for incomplete or non-standard requests.
- Functional, performance and user acceptance testing.

Out of Scope

- Direct vendor payment processing.
- Warehouse hardware installation and imaging.
- Non-standard custom laptop engineering.
- Physical delivery tracking unless a separate integration is configured.

5. Proposed Solution & Flow Design

A record-triggered Salesforce Flow is recommended for the automation. The Flow starts when a laptop request is submitted or reaches the status configured for processing.

End-to-End Process

Step	Flow Stage	Description
1	Employee submits request	Required employee, department, location, laptop model and business justification are captured.
2	Validate request	Flow checks mandatory fields, approved model, budget and duplicate request conditions.
3	Determine approval	Flow evaluates whether manager / procurement approval is required.

4	Approval	Request is routed to the appropriate approver and status is updated.
5	Create procurement record	After approval, Flow creates or updates the procurement order/request record.
6	Notify stakeholders	Employee, manager and procurement team receive status notifications.
7	Close / hand off	Request is marked Ready for Procurement or Completed based on the configured process.

Flow Logic – High-Level Design

- Start → Laptop Request created/submitted
- Get Request details → Employee / Department / Model / Cost / Status
- Decision: Are mandatory fields complete?
- No → Set status = Needs Information → Notify requester → End
- Yes → Decision: Is laptop model approved?
- No → Set status = Exception Review → Notify procurement → End
- Yes → Decision: Is duplicate active request found?
- Yes → Set status = Duplicate / Review → Notify requester → End
- No → Decision: Does cost exceed approval threshold?
- Yes → Submit for manager/procurement approval
- No → Auto-approve standard request
- Create/update Procurement Order → Notify stakeholders → End

6. Business Rules and Automation Logic

Rule ID	Rule	Implementation
BR-01	Mandatory Fields	Employee, department, location, laptop model, quantity and business justification must be present.
BR-02	Approved Model	Only laptop models marked as Active/Approved can follow the standard automated path.
BR-03	Quantity	Quantity must be a positive whole number and must not exceed the configured request limit.
BR-04	Duplicate Check	An active request for the same employee and device category should be flagged for review.
BR-05	Approval Threshold	Requests above the configured cost threshold require approval.

BR-06	Non-Standard Request	Requests outside the approved catalog are routed to Exception Review.
BR-07	Notification	Status changes generate notifications to the relevant requester, approver or procurement user.
BR-08	Auditability	The Flow records status, owner and relevant timestamps for traceability.

7. Data Model and Configuration

The following logical objects/fields can be implemented using Salesforce custom objects, standard objects, or an existing procurement data model.

Record / Object	Important Fields
Laptop Request	Request ID, Employee, Department, Location, Model, Quantity, Justification, Cost, Status
Laptop Catalog	Model, Manufacturer, Specification, Unit Cost, Active, Standard/Non-Standard
Approval	Request, Approver, Approval Level, Decision, Decision Date, Comments
Procurement Order	Order ID, Request, Vendor, Quantity, Total Cost, Order Status, Order Date
Notification Log	Request, Recipient, Notification Type, Sent Date, Delivery Status

Recommended Flow Elements

- Start / Record Trigger
- Get Records – retrieve employee, catalog and existing request data
- Assignment – prepare calculated values
- Decision – validate request and select processing path
- Create Records – create procurement request/order
- Update Records – update request and approval status
- Approval / Action – route approval where applicable
- Email Alert / Notification – communicate status
- Fault Path – capture errors and notify support

8. Functional Test Scenarios & Results

The following test cases are suitable for the functional testing template shown in the reference material. Actual results and screenshots should be updated after execution.

Test Case ID	Scenario	Test Steps	Expected Result	Sample Result	Screenshot
FT-01	Valid standard laptop request	Enter all valid required values and submit.	Request accepted; standard Flow path starts.	Pass	
FT-02	Missing mandatory field	Leave laptop model or employee field blank and submit.	Request is rejected or marked Needs Information; user receives message.	Pass	
FT-03	Invalid / non-standard model	Select a model not marked Approved.	Request is routed to Exception Review.	Pass	
FT-04	Duplicate request	Submit a second active request for the same employee/device category.	Duplicate is detected and request is flagged for review.	Pass	
FT-05	Approval required	Submit request above the configured cost threshold.	Approval request is routed to the correct approver.	Pass	
FT-06	Approval granted	Approver approves the request.	Status changes to Approved and procurement record is created/updated.	Pass	
FT-07	Approval rejected	Approver rejects the request.	Status changes to Rejected; requester is notified.	Pass	
FT-08	Notification	Trigger key status changes.	Correct stakeholders receive the expected notification.	Pass	
FT-09	Quantity validation	Enter zero, negative or excessive quantity.	Validation prevents invalid quantity and displays a clear message.	Pass	
FT-10	Flow fault handling	Simulate record creation/update failure.	Fault path captures error and alerts support; request is not silently lost.	Pass	

9. Performance Test

Performance testing should verify that the automation completes within an acceptable time and does not create excessive Salesforce processing, query, or notification overhead. The exact thresholds should be confirmed against the project's Salesforce environment and data volume.

Test ID	Scenario	Test Method	Expected Result
PT-01	Single request processing	Submit one valid standard request.	Flow completes promptly and reaches expected status.
PT-02	Concurrent requests	Submit multiple standard requests within a short period.	Requests are processed without failures or unexpected delays.

PT-03	Large catalog lookup	Process a request against a realistic catalog size.	Catalog lookup remains responsive and returns the correct model.
PT-04	Duplicate check	Run request with many historical records.	Duplicate check remains accurate without unnecessary record retrieval.
PT-05	Notification processing	Trigger multiple status notifications.	Notifications are generated once per required event.
PT-06	Fault recovery	Force a downstream record/update error.	Fault path executes and preserves useful diagnostic information.

Performance evidence to capture:

- Flow debug run duration / execution details
- Error count and failed transactions
- Record/query behavior for realistic data volume
- Notification delivery evidence

10. User Acceptance Testing (UAT)

UAT confirms that the automated laptop procurement process meets the business requirements from the perspective of employees, managers, procurement users and IT support.

UAT ID	Test Scenario	Acceptance Criteria	Sample Result	Feedback
UAT-01	Employee submits standard laptop request	Request can be submitted with clear guidance.	Pass	
UAT-02	Employee submits incomplete request	System identifies missing information.	Pass	
UAT-03	Manager approves request	Approval is easy to locate and status updates correctly.	Pass	
UAT-04	Manager rejects request	Rejection is recorded and requester is informed.	Pass	
UAT-05	Procurement reviews approved request	Procurement user can identify all required order details.	Pass	
UAT-06	Exception request	Non-standard request is clearly routed for manual review.	Pass	
UAT-07	Request status tracking	Stakeholders can see the current request status.	Pass	
UAT-08	Audit information	Relevant dates, owner and approval information are retained.	Pass	

UAT Sign-Off

Tester Name	
Business Owner	
Date	
Signature	

11. Defect Tracking and Acceptance Report

Defect Tracking Template

Bug ID	Bug Description	Steps to Reproduce	Severity	Status	Additional Feedback
BG-001	Example: duplicate request not detected	Submit two active requests for same employee/model.	Medium	Open	
BG-002	Example: notification not delivered	Trigger approval decision and verify recipient.	Low	Open	

Acceptance Summary

Section	Total	Passed	Failed	Not Tested
Functional test cases	10	10	0	0
UAT scenarios	8	8	0	0
Performance scenarios	6	6	0	0

Sample acceptance statement: The standard laptop procurement automation is considered ready for controlled deployment when all critical functional and UAT cases pass, no unresolved high-severity defects remain, and the business owner approves the final process.

12. Security, Controls and Exception Handling

Control	Implementation
Access Control	Only authorized users should create, approve, or modify procurement requests.
Least Privilege	Flow actions should operate with only the access required for the business process.
Data Validation	Use validation rules and Flow decisions to prevent incomplete or invalid requests.

Audit Trail	Maintain request status, owner, approval decision and relevant timestamps.
Fault Handling	Use Flow fault paths to capture errors and notify support rather than silently failing.
Duplicate Prevention	Check for existing active requests before creating a new procurement record.
Sensitive Data	Avoid storing unnecessary personal or financial information in the request.

13. Deployment Checklist

- ☐ Confirm approved laptop catalog and pricing are current.
- ☐ Confirm request fields and validation rules are active.
- ☐ Confirm Flow entry criteria and decision rules.
- ☐ Confirm approval routing and approver assignments.
- ☐ Confirm email templates / notifications.
- ☐ Confirm fault paths and error logging.
- ☐ Run functional regression tests in the target environment.
- ☐ Run UAT and obtain business owner sign-off.
- ☐ Document rollback / disable procedure.
- ☐ Monitor initial production transactions after deployment.

14. Conclusion

The proposed Salesforce Flow provides a structured approach to automating standard laptop procurement. By validating requests, routing approvals, creating procurement records, and notifying stakeholders, the process can reduce repetitive manual work while improving consistency, traceability and visibility.

The documentation provides a ready-to-use basis for project development, functional testing, performance testing and UAT. Project-specific screenshots, actual execution times, Salesforce Flow debug evidence, defect results and final sign-off should be added during execution.

Appendix A – Suggested Salesforce Flow Build Sequence

1. Create / confirm Laptop Request fields.
2. Create / confirm Laptop Catalog records and Approved flag.

3. Create record-triggered Flow on Laptop Request.
4. Get related employee and laptop catalog data.
5. Validate required fields and quantity.
6. Check approved model and duplicate active request.
7. Calculate total estimated cost.
8. Evaluate approval threshold.
9. Route approval when required.
10. Create/update Procurement Order after approval.
11. Update request status and timestamps.
12. Send notifications.
13. Add fault paths and error notifications.
14. Activate only after functional and UAT approval.

Appendix B – Screenshot Evidence Checklist

- Flow Builder canvas showing the complete automation.
- Laptop Request record before submission.
- Validation / error message for an invalid request.
- Approval request and approval decision.
- Created Procurement Order record.
- Notification email / Salesforce notification.
- Flow Debug Details showing a successful run.
- Fault-path or exception handling evidence.

END OF DOCUMENT